

EEE141L: ELECTRICAL CIRCUITS LAB

LAB: 07

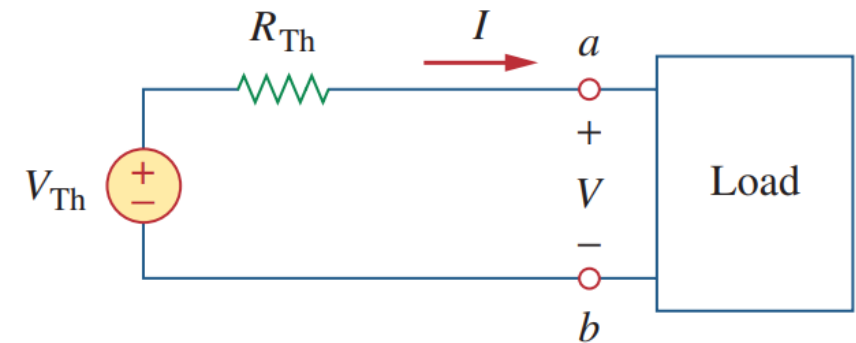
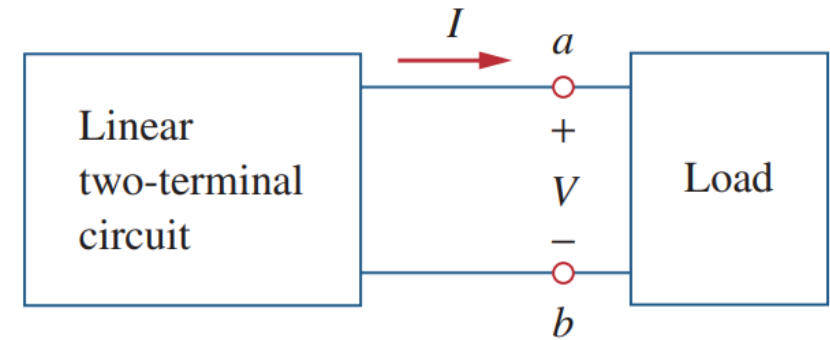
EXPERIMENT NAME: VERIFICATION OF THEVENIN'S, NORTON'S, AND
MAXIMUM POWER TRANSFER THEOREM

LAB INSTRUCTOR: SUDIPTO SARKER SUPTO

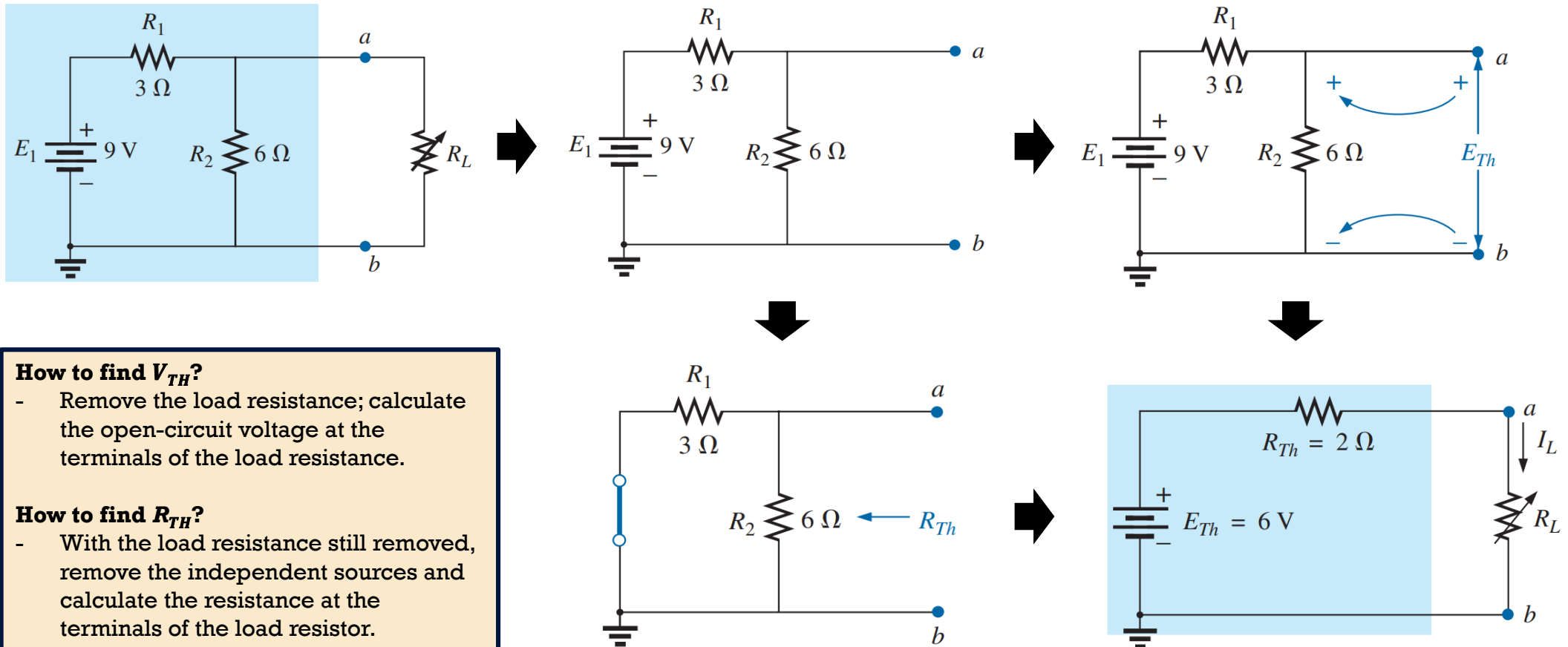


THEVENIN'S THEOREM

- Thevenin's theorem states that it is possible to simplify any linear circuit, no matter how complex, to an equivalent circuit with just a single voltage source and series resistance connected to a load.
- Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} .
- Here V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.

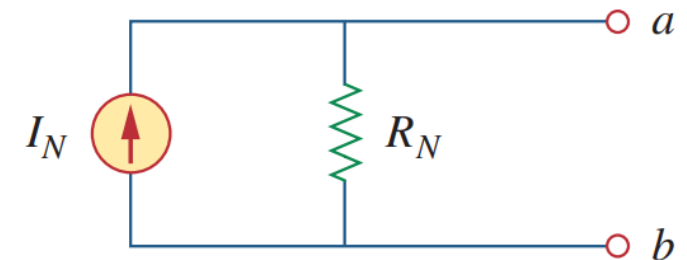
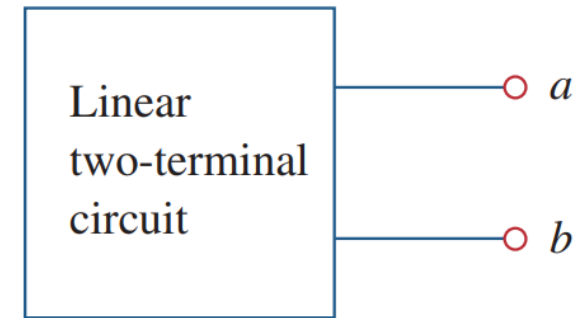


THEVENIN'S THEOREM PROCEDURE

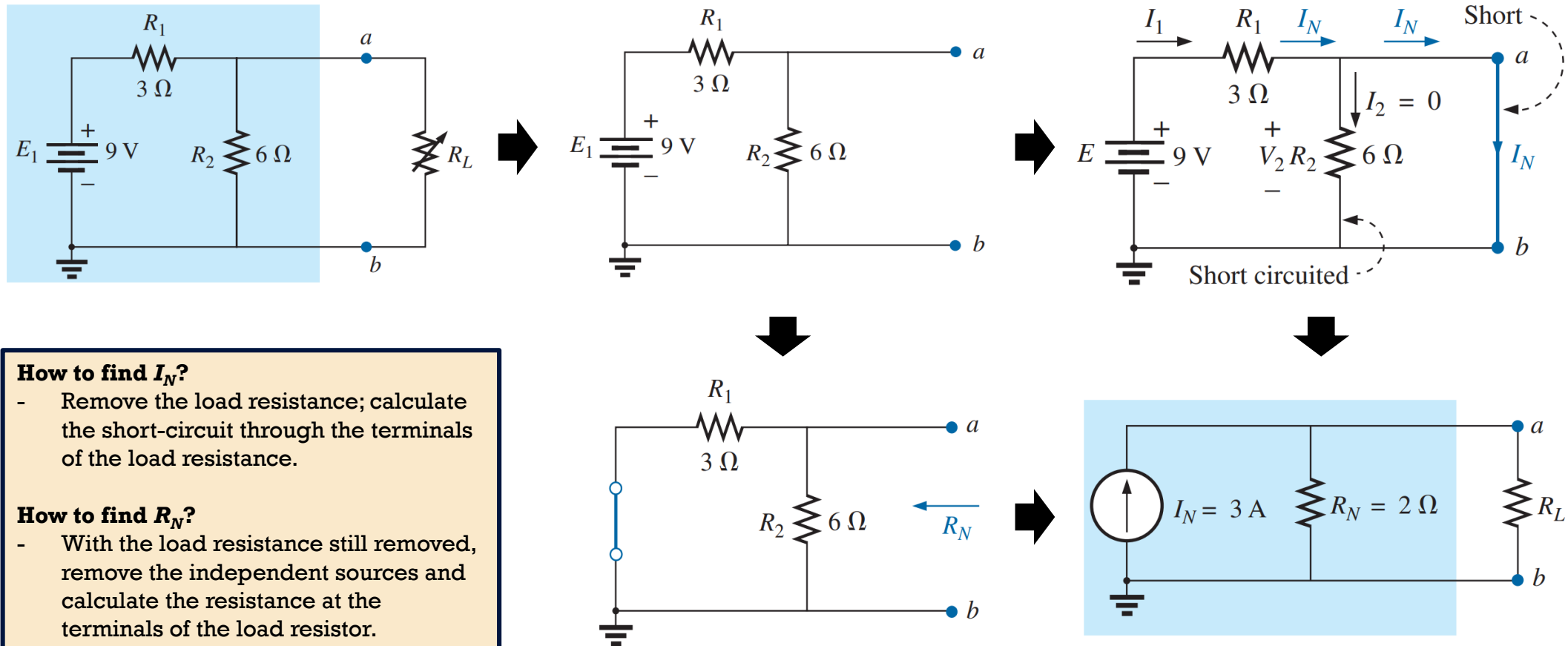


NORTON'S THEOREM

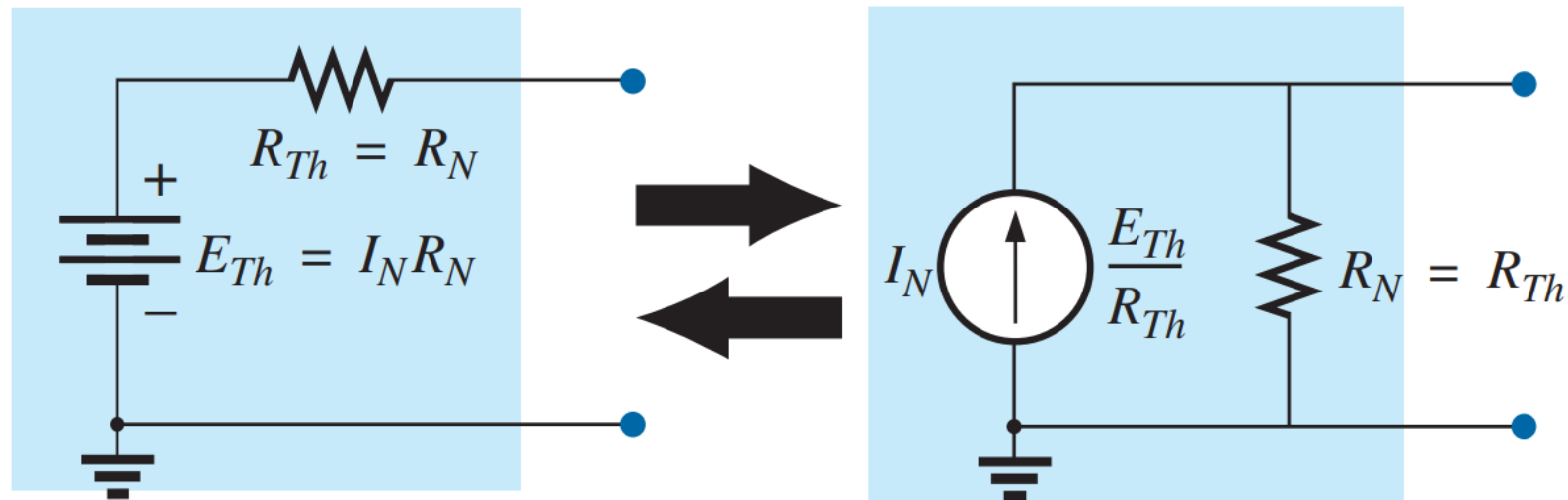
- Norton's theorem states that it is possible to simplify any linear circuit, no matter how complex, to an equivalent circuit with just a single current source and parallel resistance connected to a load.
- Norton's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N .
- Here I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.



NORTON'S THEOREM PROCEDURE



THEVENIN-NORTON EQUIVALENCE



$$R_{Th} = R_N$$
$$E_{Th} = I_N R_N$$

MAXIMUM POWER TRANSFER THEOREM

- The maximum power transfer theorem states that a load will receive maximum power from a network when its resistance is exactly equal to the Thevenin resistance of the network applied to the load. That is,

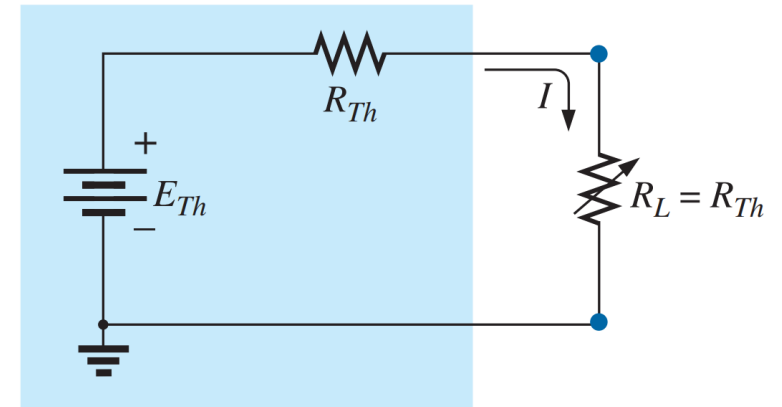
$$R_L = R_{Th}$$

- We can determine the maximum power delivered to the load by first finding the current:

$$I_L = \frac{E_{Th}}{R_{Th} + R_L} = \frac{E_{Th}}{R_{Th} + R_{Th}} = \frac{E_{Th}}{2R_{Th}}$$

- Then we substitute into the power equation:

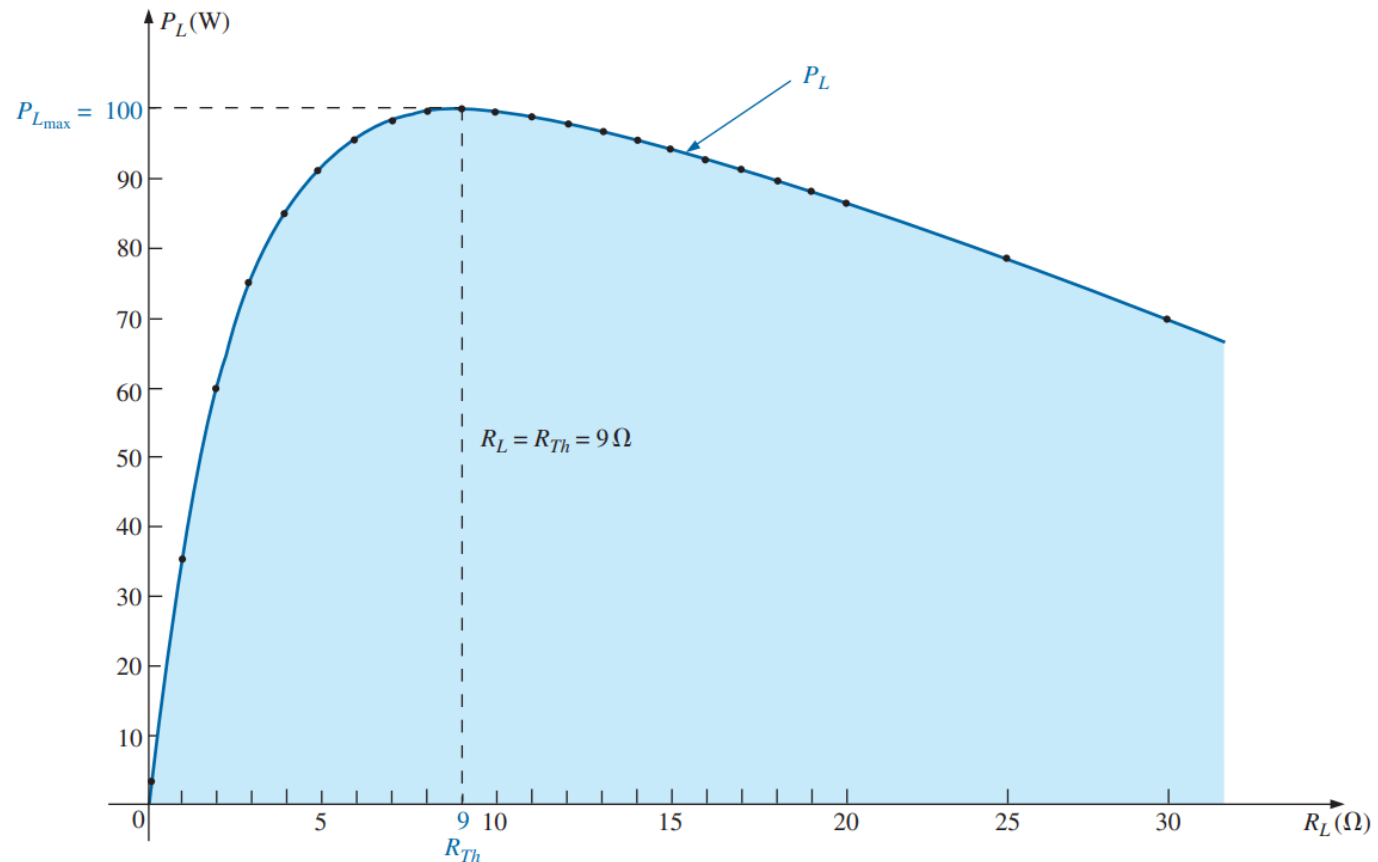
$$P_L = I_L^2 R_L = \left(\frac{E_{Th}}{2R_{Th}} \right)^2 (R_{Th}) = \frac{E_{Th}^2 R_{Th}}{4R_{Th}^2}$$



$$P_{L_{max}} = \frac{E_{Th}^2}{4R_{Th}}$$

MAXIMUM POWER TRANSFER THEOREM

CONTINUED...



Even though the load gets the maximum possible power, half of the energy is wasted in the Thevenin equivalent resistance R_{th} or source's internal resistance R_s .

THE END...

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